

Database Design Documentation for Alembic

Core System Architecture

Collaborative Design

September 29, 2025

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1 Overall Database Strategy

Our design follows the **"database per service"** model, a core tenet of microservices. This means each service owns and manages its own data, preventing tight coupling between services. It also allows for **polyglot persistence**—using the best type of database for each specific job.

- **Identity & Access Control Services:** Will use a **relational database** (e.g., PostgreSQL) to manage the structured relationships between users, organizations, and groups.
- **Knowledge Core Service:** Will use two databases: a **primary database** (like PostgreSQL or a document database like MongoDB) for the main data and a specialized **vector database** (like Pinecone or Weaviate) for the embeddings.
- **Audit Service:** Will use an **append-only log database** or a time-series database to ensure an immutable record of events.

2 Knowledge Quantum Schema (Final)

This is the central data object of the entire system, designed to be a universal container for any piece of knowledge.

QuantumID: string - *The unique identifier.*

UserID: string - *The author of the quantum.*

OrganisationID: string - *The organization it belongs to.*

Visibility: string - *Access switch: 'private', 'organisation', 'groups'.*

AccessGroupIDs: array[string] - *Which groups can access it if visibility is 'groups'.*

Timestamp: datetime - *When it was created/last updated.*

Topic: string - *A high-level, human-defined category or project.*

CoreInsight: string - *The central, human-readable piece of knowledge.*

Embeddings: array[object] - *An array for model-independent semantic data.*

- ModelID: string

- Vector: array[float]

SupportingFacts: array[string] - *A list of raw text evidence.*

SupportingQuanta: array[QuantumID.Reference] - *Links to other quanta, forming the knowledge graph.*

ReasoningChain: string - *The narrative that connects the evidence to the insight.*

ConfidenceScore: float - *A score from 0.0 to 1.0 representing confidence.*

Tags: array[string] - *Specific, human-defined keywords for filtering.*

SourceID: string - *A reference to the original conversation or document.*

3 Audit Log Entry Schema

This schema is designed to be lean and secure, providing a complete audit trail without duplicating sensitive information.

LogID: `string` - *The unique identifier for the log entry.*

Timestamp: `datetime` - *When the event occurred.*

UserID: `string` - *The user who performed the action.*

Action: `string` - *The type of event (e.g., CREATE, UPDATE, DELETE, ACCESS).*

TargetType: `string` - *The type of object affected (e.g., KnowledgeQuantum, User).*

TargetID: `string` - *The unique ID of the object affected.*

Changes: `array[string]` - *Optional: For UPDATE actions, lists which fields were changed, but not the new content itself.*